

Akhil M M

Mern Stack Developer

☎ +91-9778005280 | ✉ manojakhil36@gmail.com | 🐙 [GitHub](#) | 💼 [LinkedIn](#) | 🌐 [Portfolio](#) | 📄 [LeetCode](#)

Profile Summary

Full Stack Developer skilled in building scalable web applications and backend systems using TypeScript, Node.js, Express.js, React.js, and MongoDB. Experienced in designing real-time applications, integrating third-party APIs, and deploying production-ready systems with a focus on performance and maintainability.

Skills

Languages: TypeScript, JavaScript (ES6+)

Frontend: React.js, Next.js, Redux Toolkit, Tailwind CSS, Material UI, HTML, CSS

Backend: Node.js, Express.js, REST APIs, Socket.IO, JWT Authentication

Architecture: REST API Design, Repository Pattern, MVC Architecture, Authentication & Authorization

Database & Caching: MongoDB, Redis, MongoDB Atlas Vector Search

Cloud & DevOps: Docker, AWS (EC2, S3), Google Cloud Platform (GCP), NGINX, PM2, Vercel

Integrations: Stripe, Razorpay, Google Maps API, Cloudinary, WebRTC

Tools: Git, GitHub, Postman

Projects

Sarathi – Driver-On-Demand Platform

[Live](#) | [GitHub](#)

A production-grade driver-on-demand platform featuring automated driver matchmaking, secure payments, and real-time chat and video communication.

Tech: TypeScript, Node.js, Express.js, React.js, MongoDB, Redis, Socket.IO, WebRTC, Stripe, AWS EC2, NGINX

- Architected the backend using **TypeScript**, **Clean Architecture**, **Tsyringe dependency injection**, and **SOLID principles** to build a maintainable and scalable codebase.
- Built a real-time driver matchmaking system using **Socket.IO** and **Google Maps API** to broadcast ride requests and assign nearby approved drivers based on road-distance calculations.
- Developed peer-to-peer chat and video calling features using **WebRTC** and **Socket.IO**, enabling real-time messaging and communication between users and drivers.
- Integrated **Stripe Connect** for driver onboarding, split-payment processing, and secure transaction handling within the booking workflow.
- Deployed the platform on **AWS EC2** using **PM2** and **NGINX**, hosted the frontend on **Vercel**, and integrated **Cloudinary** for secure media storage and delivery.

AI-Powered Real-Time Group Chat Platform – Intelligent Chat Application

[Live](#) | [GitHub](#)

An AI-powered group chat platform featuring context-aware conversations, real-time messaging, and scalable architecture.

Tech: TypeScript, Node.js, Express.js, React.js, MongoDB Atlas Vector Search, Google Gemini API, Socket.IO, TanStack Virtual, AWS S3, Google Cloud Platform

- Integrated **Google Gemini API** with a custom **Retrieval-Augmented Generation (RAG)** pipeline using **MongoDB Atlas Vector Search** and Gemini embeddings to deliver context-aware AI responses.
- Architected the backend using **TypeScript** and **Clean Architecture** with modular Use Cases, Repositories, Services, and Controllers for maintainability and scalability.
- Developed real-time messaging features using **Socket.IO**, including AI-assisted conversations, read receipts, intelligent message grouping, media sharing, and event-driven communication.
- Implemented high-performance chat rendering using **TanStack Virtual**, infinite scrolling, and minimized re-renders to efficiently handle large chat histories.
- Implemented **JWT access/refresh token rotation**, **Google OAuth 2.0** authentication, **AWS S3 presigned uploads**, and deployed backend services on **Google Cloud Platform (GCP)**.

EYEGAZE – Full-Stack E-Commerce Platform

Live | [GitHub](#)

A full-stack eyewear e-commerce platform featuring secure payments, inventory management, analytics dashboards, and store administration tools.

Tech: Node.js, Express.js, MongoDB, EJS, Bootstrap, Chart.js, Razorpay

- Developed secure authentication and authorization workflows using **session-based authentication** and custom middleware for role-based access control.
- Integrated the **Razorpay payment gateway** to enable secure checkout and order processing.
- Built an admin dashboard with **Chart.js** to visualize revenue, sales performance, and user growth analytics.
- Implemented product, category, inventory, offer, and coupon management modules to streamline store administration.
- Designed responsive user interfaces using **EJS, Bootstrap**, HTML, and CSS, and optimized multi-image uploads using **Multer** for efficient catalog management.

Experience

Venkys Group

Marketing Assistant / Sales Supervisor

Kerala, India
Jul 2022 – May 2024

- Managed customer orders and coordinated with internal teams to support sales and day-to-day operations.
- Handled customer interactions and ensured timely order fulfillment across different regions.

Education

Professional Training – Full Stack Development (MERN Stack)

Prototype

Jul 2024 – May 2026

- Built and deployed full-stack web applications using React.js, Node.js, Express.js, and MongoDB
- Developed scalable REST APIs, authentication systems, and database-driven applications
- Implemented payment gateway integrations, third-party APIs, and real-time application features
- Worked on production-style projects following modern software development practices
- Strengthened skills in debugging, performance optimization, version control, and clean code principles

B.Sc. Mathematics with Complementary Studies in Computer Science

University of Calicut

2019 – 2022
Kerala, India

Languages

English | Tamil